Classification

Binary Classification Problems

Classification: Predicting whether an observation is in a certain category. That is, we are working with a categorical response variable.

Binary Classification: Only two possible categories

Eg. “Is it going to rain today?”

Evaluating Binary Classification Models

One class of metrics is based on comparing the actual and predicted outcomes.

	Date	rained	prediction
0	2024-01-01	no rain	no rain
1	2024-01-02	no rain	rain
2	2024-01-03	no rain	no rain
3	2024-01-04	no rain	no rain
4	2024-01-05	rain	rain
5	2024-01-06	rain	rain
6	2024-01-07	no rain	rain
7	2024-01-08	rain	rain
8	2024-01-09	rain	rain
9	2024-01-10	no rain	no rain

Metrics for Classification

We can understand a lot of metrics through a **confusion matrix** which shows the actual labels vs. predicted labels.

		Predicted	
		negative	positive
Actual	negative	True Negatives (TN)	False Positives (FP)
	positive	False Negatives (FN)	True Positives (TP)

Metrics for Classification

We can understand a lot of metrics through a **confusion matrix** which shows the actual labels vs. predicted labels.

		Predicted	
		no rain	rain
Actual	no rain	148	87
	rain	43	88

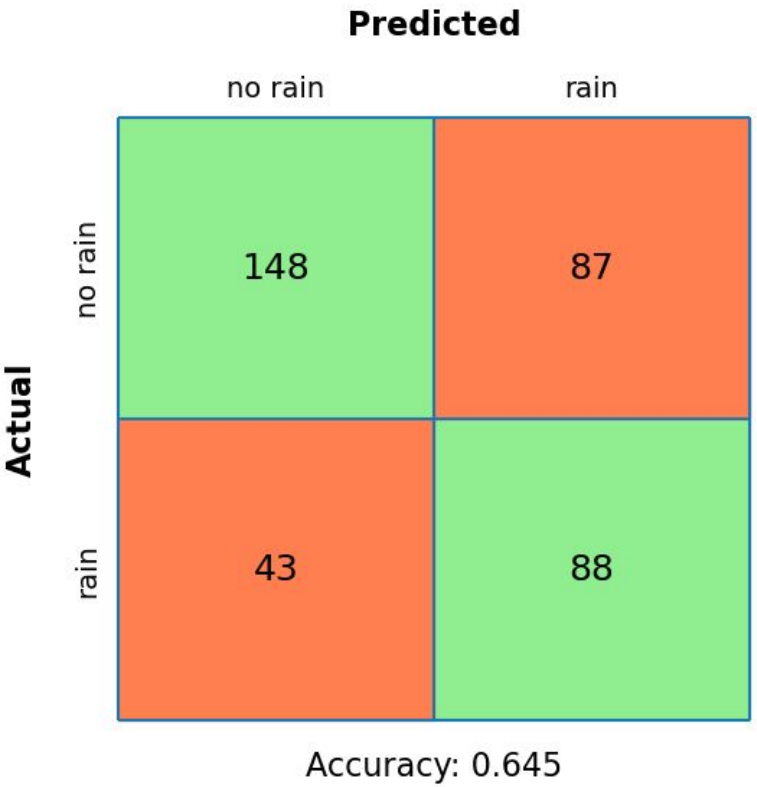
Metrics for Classification

Accuracy: What percentage of the time were the predictions correct?

$$\text{accuracy} = \frac{\text{correct predictions}}{\text{total predictions}}$$

$$\text{accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

Metrics for Classification



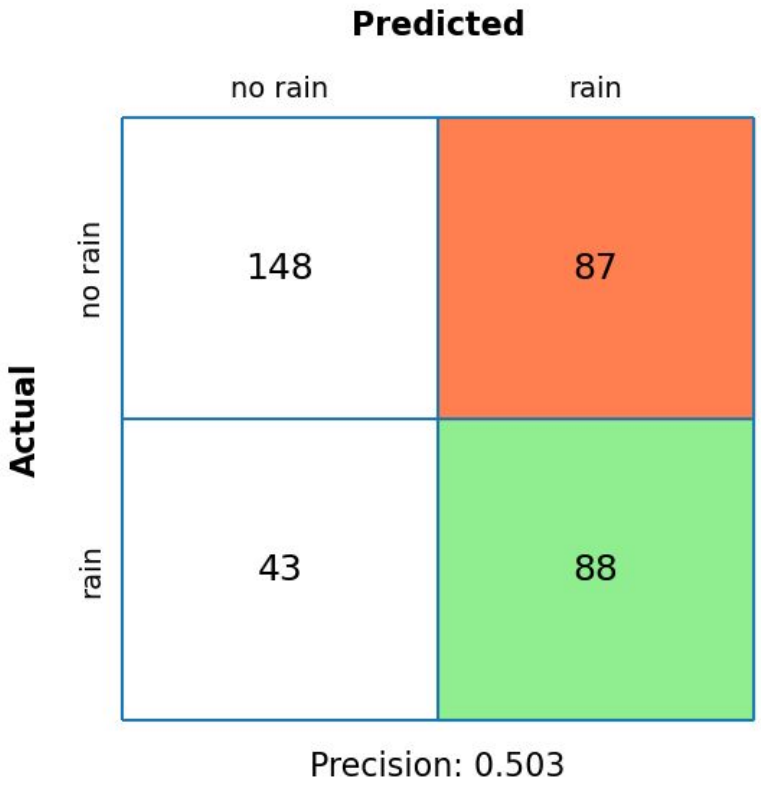
$$\text{accuracy} = \frac{\text{correct predictions}}{\text{total predictions}}$$

Metrics for Classification

Precision: Out of those predicted to be in the positive class, what percentage actually are?

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

Metrics for Classification



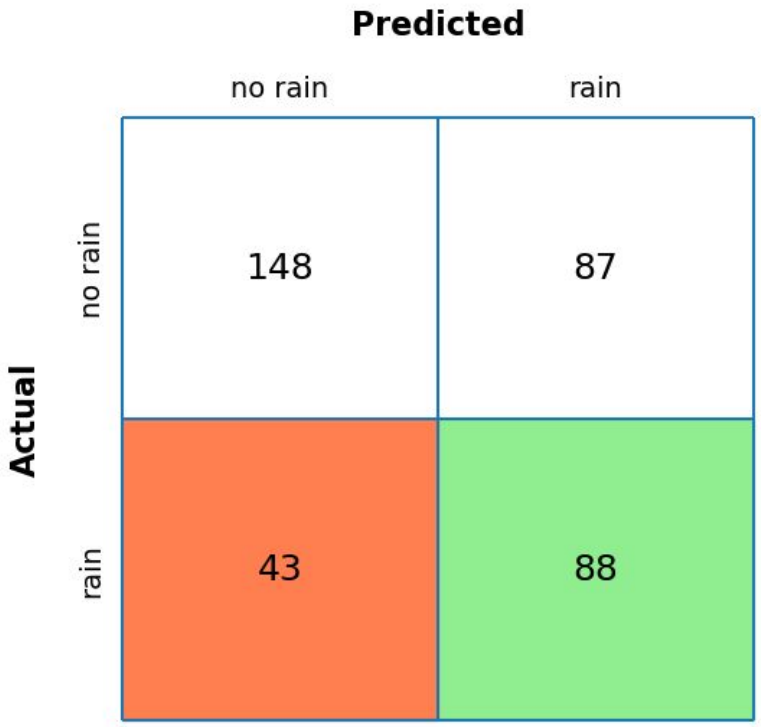
$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

Metrics for Classification

Recall: Out of those observation actually in the positive class, what percentage actually are predicted correctly?

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

Metrics for Classification



Recall: 0.672

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

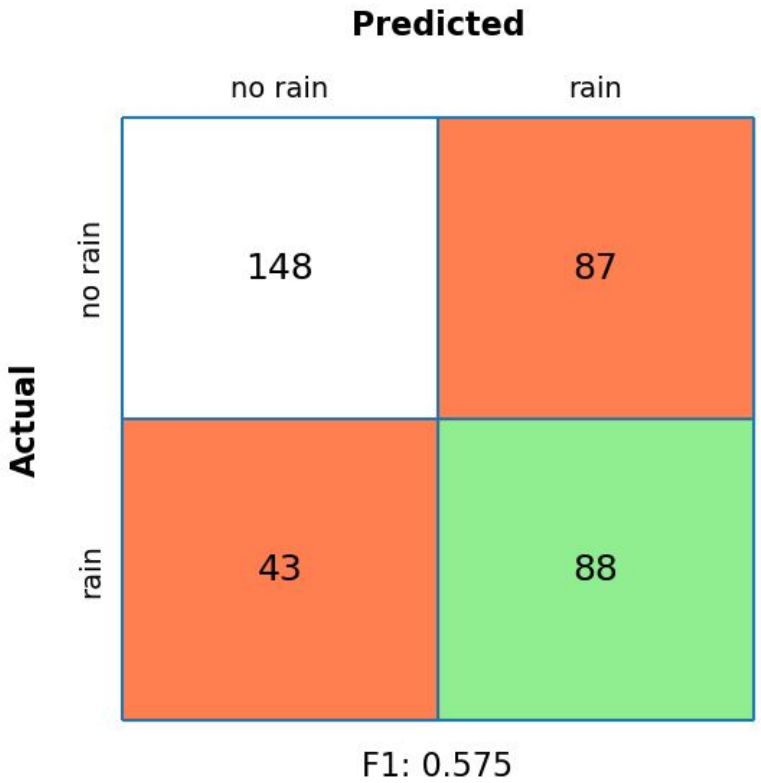
Metrics for Classification

F1 Score: The (harmonic) mean of precision and recall.

$$\mathrm{F1} = \frac{2 \cdot \mathrm{TP}}{2 \cdot \mathrm{TP} + \mathrm{FP} + \mathrm{FN}} = 2 \cdot \frac{\mathrm{precision} \cdot \mathrm{recall}}{\mathrm{precision} + \mathrm{recall}}$$

To have a good F1 score, a model must have a good precision and recall score.

Metrics for Classification



$$\mathrm{F1} = \frac{2 \cdot \mathrm{TP}}{2 \cdot \mathrm{TP} + \mathrm{FP} + \mathrm{FN}}$$

Binary Classification Problems

Some types of models will just produce a prediction, but just guessing “yes” or “no” is not necessarily the best approach, especially if there is no perfect rule.

Another approach is to predict a probability rather than just a label.

This predicted probability is often translated to a prediction by setting a threshold value.

ROC Curve

The receiver operating characteristic curve (ROC curve) looks at the relationship between the **true positive rate** and **false positive rate** as we adjust the threshold value.

$$TPR = TP / (TP + FN)$$

$$FPR = FP / (TN + FP)$$

ROC Curve

Threshold = 0.5

TPR: 0.672

FPR: 0.370

Actual \ Predicted	no rain	rain
no rain	148	87
rain	43	88

ROC Curve

Threshold = 0.75

TPR: 0.344

FPR: 0.191

Higher thresholds lead to lower false positives but also lower true positives.

		Predicted	
		no rain	rain
Actual	no rain	190	45
	rain	86	45

ROC Curve

Threshold = 0.25

TPR: 0.817

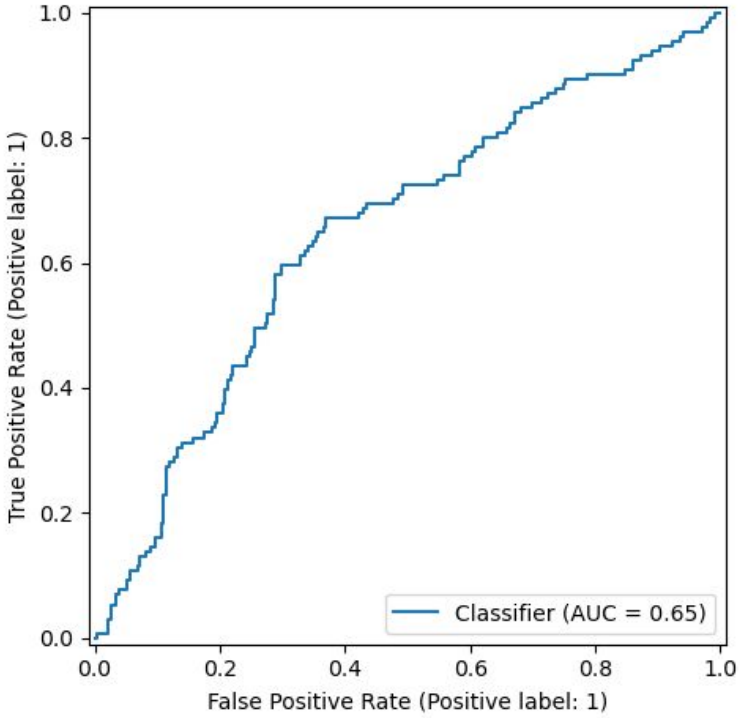
FPR: 0.664

Lower thresholds lead to higher true positives but also higher false positives.

		Predicted	
		no rain	rain
Actual	no rain	79	156
	rain	24	107

ROC Curve

Plotting out the TPR and FPR over all possible thresholds gives the **receiver operating characteristic curve.**

We can calculate the area under this curve. The closer to 1, the better.

A high area means that the model tends to assign higher probability to observations in the positive class than to the negative class.